Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim: Image Processing 3**	
Experiment -9	**Date:25/08/2026**	**Enrollment No:92400133057**

Github Link :- https://github.com/manavkasundra/DSIP

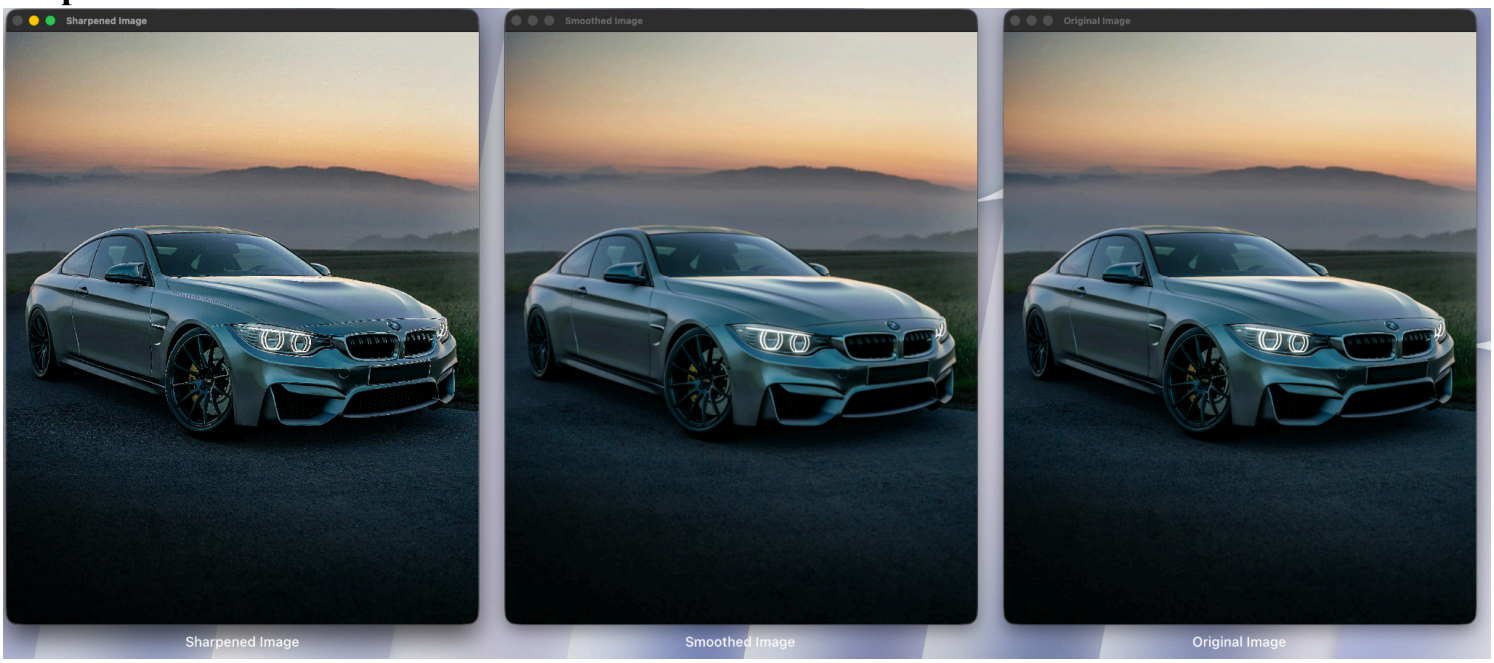
Code :-

1. python program to simulate smoothening and sharpening operation on image using spatial filter.

```
import cv2
import numpy as np
image = cv2.imread('img.jpg')
if image is None:
   print("Error: Could not load img.jpg")
   exit()
kernel_size = (5, 5)
sigma = 1.5
gaussian_1d = cv2.getGaussianKernel(5, sigma)
gaussian_kernel = np.outer(gaussian_1d, gaussian_1d)
smoothed_image = cv2.filter2D(image, -1, gaussian_kernel)
sharpening_kernel = np.array([
   [-1, -1, -1],
   [-1,  9, -1],
   [-1, -1, -1]
])
sharpened_image = cv2.filter2D(image, -1, sharpening_kernel)
cv2.imshow("Original Image", image)
cv2.imshow("Smoothed Image", smoothed_image)
cv2.imshow("Sharpened Image", sharpened_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

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Output:-

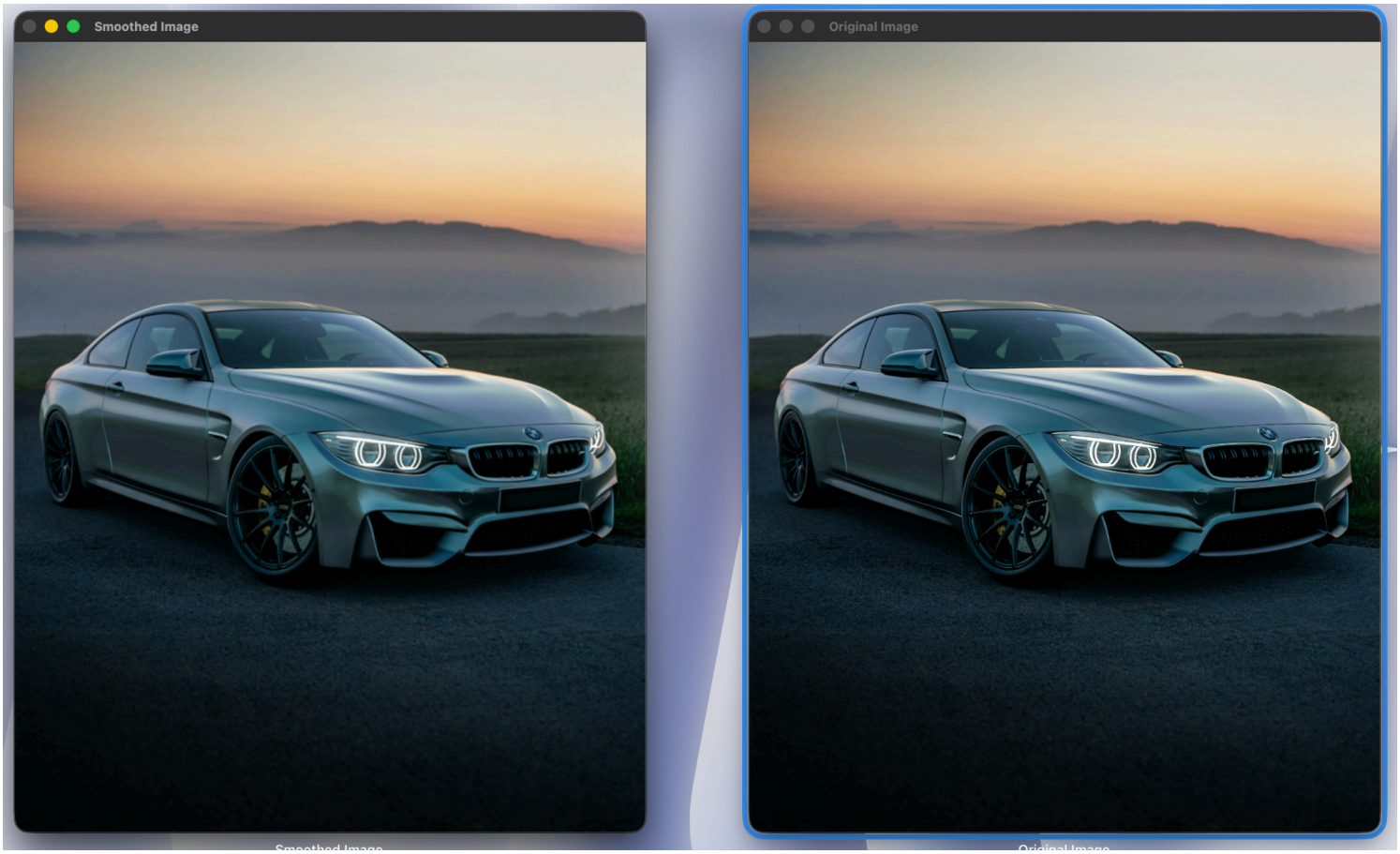


2. python program to simulate smoothening using Averaging linear filters

```
import cv2
import numpy as np
image = cv2.imread('img.jpg')
if image is None:
   print("Error: Could not load img.jpg")
   exit()
kernel_size = (5, 5)
kernel = np.ones(kernel_size, dtype=np.float32) / (
   kernel_size[0] * kernel_size[1]
)
smoothed_image = cv2.filter2D(image, -1, kernel)
cv2.imshow("Original Image", image)
cv2.imshow("Smoothed Image", smoothed_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:-

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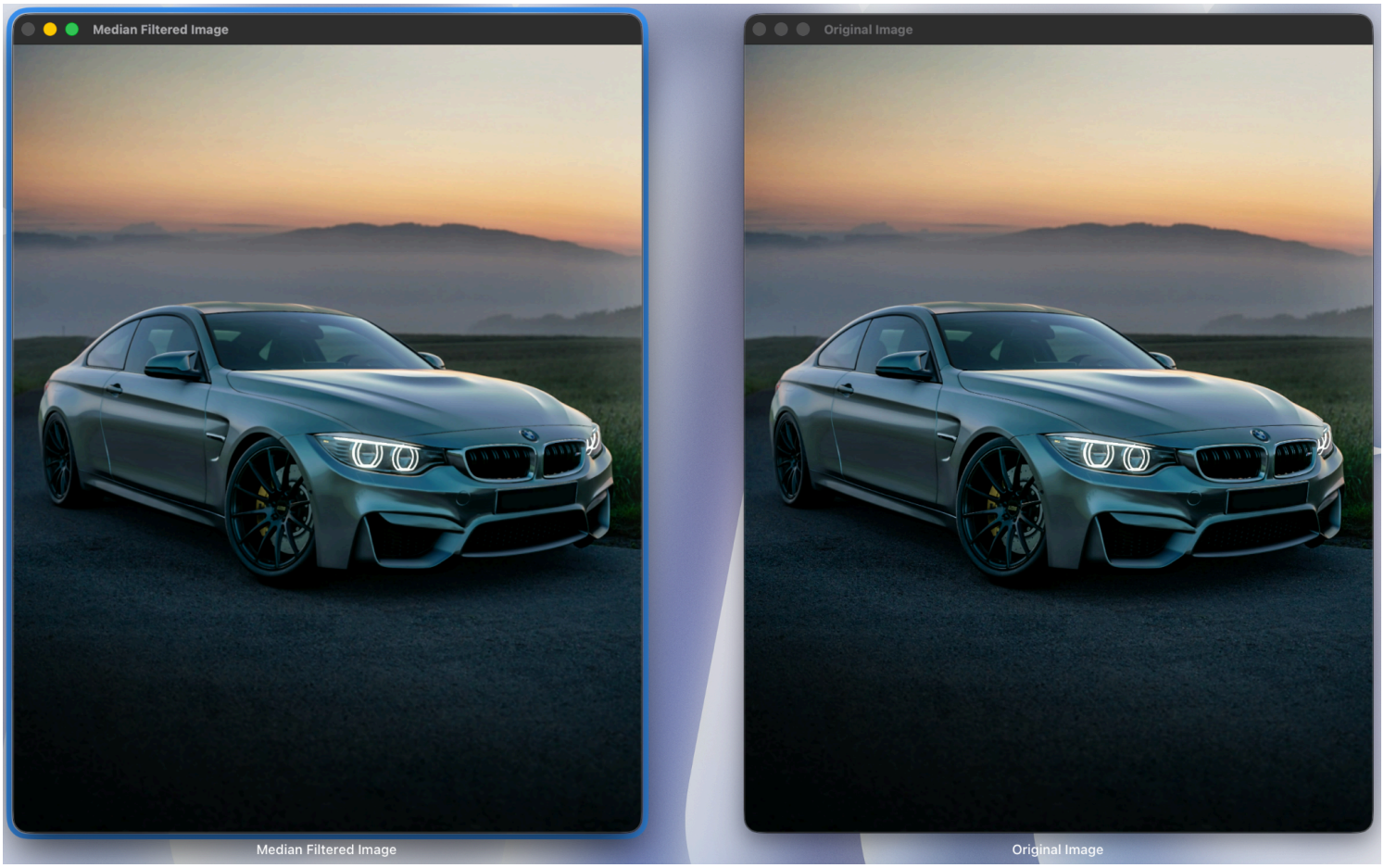


3. python program to simulate smoothening using median filters

```
import cv2
import numpy as np
image = cv2.imread('img.jpg')
if image is None:
   print("Error: Could not load img.jpg")
   exit()
kernel_size = 5
smoothed_image = cv2.medianBlur(image, kernel_size)
cv2.imshow("Original Image", image)
cv2.imshow("Median Filtered Image", smoothed_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:-

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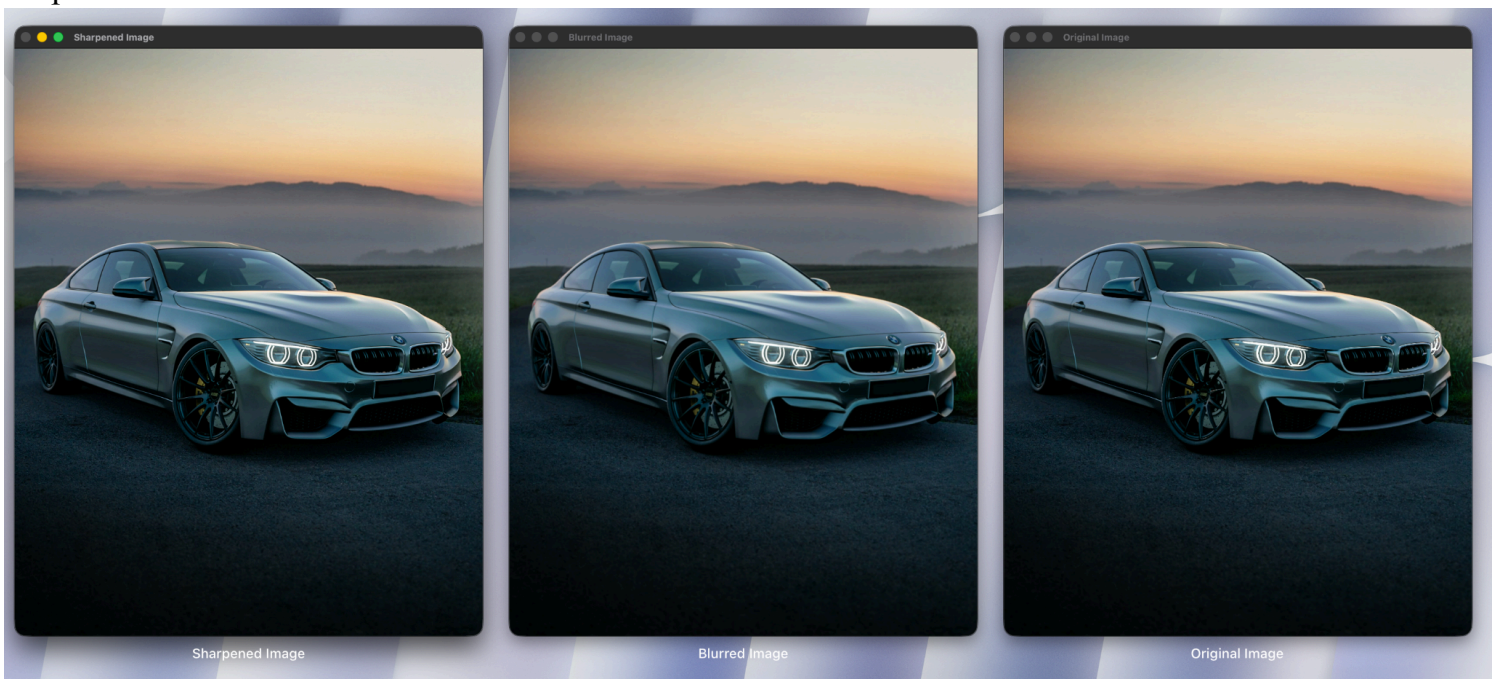


4. python program to simulate sharpening using spatial high pass filters

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```python
import cv2
import numpy as np
image = cv2.imread('img.jpg')
if image is None:
   print("Error: Could not load img.jpg")
   exit()
blurred_image = cv2.GaussianBlur(image, (5, 5), 0)
laplacian_kernel = np.array([
   [0, -1,  0],
   [-1,  5, -1],
   [0, -1,  0]
], dtype=np.float32)
sharpened_image = cv2.filter2D(
   blurred_image, -1, laplacian_kernel
)
cv2.imshow("Original Image", image)
cv2.imshow("Blurred Image", blurred_image)
cv2.imshow("Sharpened Image", sharpened_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:-



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Conclusion:-

In this experiment, we successfully explored the fundamentals of spatial filtering in image processing using the OpenCV library. By applying different kernel-based operations, we observed how specific mathematical transformations directly impact image characteristics:

Smoothing and Noise Reduction: By utilizing averaging and Gaussian filters, we were able to reduce high-frequency noise and soften the overall texture of the image, demonstrating the practical application of blurring techniques for preprocessing. Additionally, we found that median filtering offers a distinct advantage in effectively removing "salt-and-pepper" noise while maintaining sharper boundaries than linear filters.

Edge Enhancement: Through the application of high-pass filters, such as the Laplacian kernel, we successfully accentuated the fine details and edges within the image. This process clarified structural features that were previously subtle, proving the effectiveness of sharpening filters in image post-processing.

Overall, this experiment highlights the versatility of spatial kernels as a foundational tool for image enhancement. By carefully selecting the appropriate filter, one can achieve a balance between noise reduction and detail preservation, which is critical for various computer vision applications.